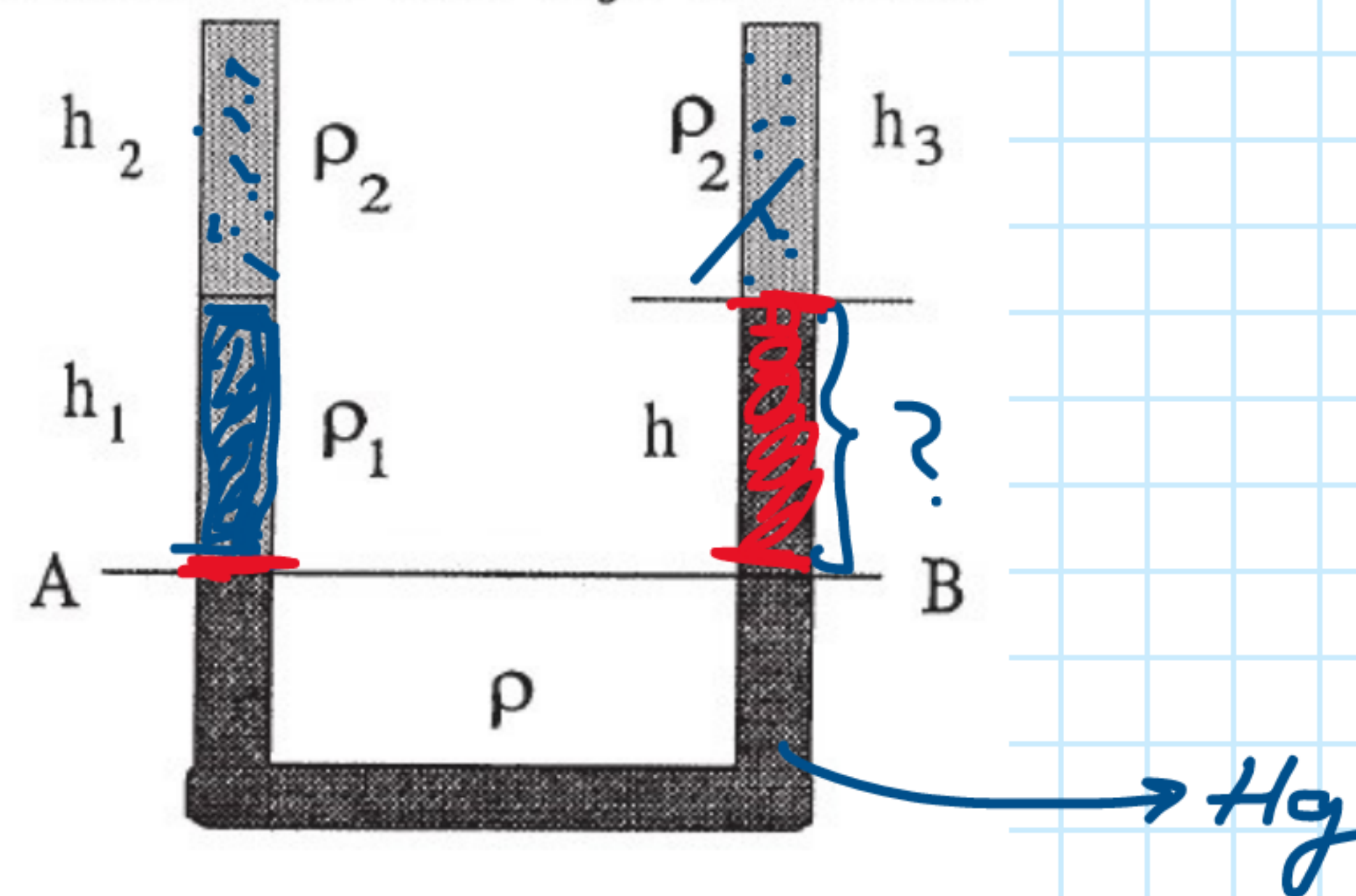


Ushtrime ND8b

1.2.113 Në gypin e formës U me krah të njejtë e vendosim pak mërkur, pastaj në njërin krah e vendosim pak ujë në lartësi $h=20\text{cm}$, e pastaj i mbushim tërësisht me vaj. Sa është ndryshimi i nivelit të merkurit në të dy krahët.

Zgjidhje:



Të dhënat:

$$h_1 = 20\text{cm} = 20 \cdot 10^{-2}\text{m}$$

$$\rho_1 = 1000 \frac{\text{kg}}{\text{m}^3}$$

$$\rho = 13000 \frac{\text{kg}}{\text{m}^3}$$

$$\rho_2 = 900 \frac{\text{kg}}{\text{m}^3}$$

$$h = ?$$

$$\cancel{\rho_1} \rho_1 + \rho_2 = \rho + \rho_2 + \cancel{\rho_2}$$

$$\rho_1 g h_1 + \rho_2 g h_2 = \rho g h + \rho_2 g h_3 \quad | :g$$

$$\rho_1 h_1 + \rho_2 h_2 = \rho h + \rho_2 h_3 \quad *$$

$$h_1 + h_2 = h + h_3 \Rightarrow \boxed{h_3 - h_2 = h_1 - h} \quad \text{nga figura.}$$

$$\rho_1 h_1 - \rho h = \rho_2 h_3 - \rho_2 h_2$$

$$\rho_1 h_1 - \rho h = \rho_2 (h_3 - h_2)$$

$$\rho_1 h_1 - \rho h = \rho_2 (h_1 - h)$$

$$\rho_1 h_1 - \rho h = \rho_2 h_1 - \rho_2 h$$

$$\rho_1 h_1 - \rho_2 h_1 = -\rho h + \rho_2 h$$

$$(\rho_1 - \rho_2) h_1 = (\rho - \rho_2) h \quad | : (\rho - \rho_2)$$

$$h = h_1 \frac{\rho_1 - \rho_2}{\rho - \rho_2}$$

$$h = 20 \cdot 10^{-2}\text{m} \frac{1000 \frac{\text{kg}}{\text{m}^3} - 900 \frac{\text{kg}}{\text{m}^3}}{13000 \frac{\text{kg}}{\text{m}^3} - 900 \frac{\text{kg}}{\text{m}^3}}$$

$$h = 20 \cdot 10^{-2}\text{m} \frac{100 \frac{\text{kg}}{\text{m}^3}}{12700 \frac{\text{kg}}{\text{m}^3}} = \frac{20}{127} \cdot 10^{-2}\text{m} = 0,157 \cdot 10^{-2}\text{m}$$

$$h = 1,57 \cdot 10^{-3}\text{m} = \underline{\underline{1,57\text{mm}}}$$